

DBMS –SELLER AND INVENTORY TABLE

Flipkart Seller and Inventory Management System Using Oracle SQL

1.Database Design

1.1 Seller Table

Entity: Seller

Attribute	Key type
Seller_ID	Primary key
Seller_Name	Not Null
Store_name	Unique key
Email	Unique key
Phone_number	-

1.2 Inventory Table

Inventory ID	Product ID	Seller ID	Stock Quantity	Stock Status	Last Updated
301	101	201	20	Available	17-SEP-26
302	102	202	30	Available	17-SEP-26
303	103	203	50	Available	17-SEP-26
304	104	201	15	Available	17-SEP-26
305	105	204	0	Unavailable	17-SEP-26
306	106	205	40	Available	17-SEP-26

2.Table Creation

2.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(50) NOT NULL,  
4     Store_Name VARCHAR2(100),  
5     Email VARCHAR2(100) UNIQUE,  
6     Phone_Number VARCHAR2(15)  
7 );  
  
Table created.
```

2.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
  2     Inventory_ID NUMBER PRIMARY KEY,  
  3     Product_ID NUMBER,  
  4     Seller_ID NUMBER,  
  5     Stock_Quantity NUMBER,  
  6     Stock_Status VARCHAR2(20),  
  7     Last_Updated DATE,  
  8     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID),  
  9     FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID)  
10 );  
  
Table created.
```

3.Seller Data Insertion

```
SQL> INSERT INTO Seller VALUES  
  2  (201, 'Arun Kumar', 'Arun Electronics', 'arunstore@gmail.com', '9876500001');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller VALUES  
  2  (202, 'Priya Sharma', 'Priya Fashion', 'priyastore@gmail.com', '9876500002');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller VALUES  
  2  (203, 'Kavin Raj', 'Kavin Stores', 'kavinstore@gmail.com', '9876500003');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller VALUES  
  2  (204, 'Divya S', 'Divya Books', 'divyastore@gmail.com', '9876500004');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller VALUES  
  2  (205, 'Rahul Kumar', 'Rahul Beauty', 'rahulstore@gmail.com', '9876500005');  
  
1 row created.  
  
SQL>  
SQL> COMMIT;
```

4.Inventory Data Insertion

```
SQL> INSERT INTO Inventory VALUES (301, 101, 201, 20, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (302, 102, 202, 30, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (303, 103, 203, 50, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (304, 104, 201, 15, 'Available', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (305, 105, 204, 0, 'Unavailable', SYSDATE);
1 row created.

SQL>
SQL> INSERT INTO Inventory VALUES (306, 106, 205, 40, 'Available', SYSDATE);
1 row created.
```

5. Seller Product Information

5.1 Display Seller and Product Details

This query combines Seller, Inventory, and Product information

```

SQL> SELECT
2     s.Seller_Name,
3     p.Product_Name,
4     p.Price,
5     p.Stock
6 FROM Seller s
7 JOIN Product p
8 ON s.Seller_ID = p.Seller_ID;

```

SELLER_NAME

PRODUCT_NAME

PRICE	STOCK
Arun Kumar Laptop	55000 20

Priya Sharma Smartphone	25000 30
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SELLER_NAME

PRODUCT_NAME

PRICE	STOCK
Kavin Raj T-Shirt	799 50

Arun Kumar Mixer Grinder	
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SELLER_NAME

PRODUCT_NAME

PRICE	STOCK
Divya S Python Programming Book	650 0

Rahul Kumar

SELLER_NAME

PRODUCT_NAME

PRICE	STOCK
Face Wash	350 40

6 rows selected.

6.Inventory Status

6.1 Display Available Products

```
SQL> SELECT
  2     Inventory_ID,
  3     Product_ID,
  4     Stock_Quantity,
  5     Stock_Status
  6 FROM Inventory
  7 WHERE Stock_Quantity > 0;
```

INVENTORY_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
301	101	20	Available
302	102	30	Available
303	103	50	Available
304	104	15	Available
306	106	40	Available

6.2 Display Unavailable Products

```
SQL> SELECT
  2     Inventory_ID,
  3     Product_ID,
  4     Stock_Quantity,
  5     Stock_Status
  6 FROM Inventory
  7 WHERE Stock_Quantity = 0;
```

INVENTORY_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
305	105	0	Unavailable

7.Inventory Update

```
SQL> UPDATE Inventory
  2 SET Stock_Quantity = 25,
  3     Stock_Status = 'Available',
  4     Last_Updated = SYSDATE
  5 WHERE Inventory_ID = 301;

1 row updated.
```

8.Inventory Status Reports

8.1 Complete Inventory Report

```

SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Product_Count,
  4     SUM(Stock_Quantity) AS Total_Stock
  5 FROM Inventory
  6 GROUP BY Stock_Status;

```

STOCK_STATUS	PRODUCT_COUNT	TOTAL_STOCK
Available	5	160
Unavailable	1	0

8.2 Available and Unavailable Product Count

```

SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Product_Count
  4 FROM Inventory
  5 GROUP BY Stock_Status
  6 ORDER BY Stock_Status;

```

STOCK_STATUS	PRODUCT_COUNT
Available	5
Unavailable	1

8.3 Total Stock by Seller

```

SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     SUM(i.Stock_Quantity) AS Total_Stock
  5 FROM Seller s
  6 JOIN Inventory i
  7 ON s.Seller_ID = i.Seller_ID
  8 GROUP BY s.Seller_ID, s.Seller_Name
  9 ORDER BY s.Seller_ID;

```

SELLER_ID	SELLER_NAME	TOTAL_STOCK
201	Arun Kumar	40
202	Priya Sharma	30
203	Kavin Raj	50
204	Divya S	0
205	Rahul Kumar	40

8.4 Products with Zero Stock

```
SQL> SELECT
  2     i.Inventory_ID,
  3     i.Product_ID,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM Inventory i
  8 JOIN Seller s
  9 ON i.Seller_ID = s.Seller_ID
 10 WHERE i.Stock_Quantity = 0;
```

INVENTORY_ID	PRODUCT_ID	SELLER_NAME
305	105	Divya S
0		Unavailable